

High incidence of multi-drug resistance and heterogeneity of mobile genetic elements in *Escherichia coli* isolates from diseased ducks in Sichuan province of China

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ABSTRACT

Harmonious ecological environment is a major concern with rising feeding and consumption of ducks, as these waterfowl birds can promote the spread of antibiotic resistant genes (ARGs). Therefore, this study was conducted to know diversity of antimicrobial resistance (AMR), integrons, and mobile genetic elements (MGEs) in *Escherichia coli* (*E. coli*) isolated from intestinal contents or pericardial effusion of diseased ducks from 2018 to 2020 in Sichuan, China. The AMR phenotype was determined via disk diffusion test in 165 *E. coli* isolates. Further, the integrase genes of integron (*intI1*, *intI2* and *intI3* genes), gene cassettes (GCs) and MGEs were screened by PCR and sequencing. The results indicated 100% isolates were resistant to at least one antibiotic and 98.8% were multidrug-resistant strains. Highest AMR phenotype was recorded to rifampin (97.0%) followed by ampicillin (95.8%), chloramphenicol (89.7%), trimethoprim-sulfamethoxazole (84.2%), ciprofloxacin (83.0%), cefotaxime (80.0%), streptomycin (75.8%), doxycycline (49.7%), amikacin (10.3%), amoxicillin/clavulanic acid (3.6%), polymyxin B (1.2%) and ertapenem (0.6%). Further, class 1 and 2 integrons were found in 87.3% and 17.6% isolates, respectively. All isolates were negative for *intI3* gene. The variable region of class 1 and 2 integrons contained total 13 different GCs, including *arr-3*+*dfrA27*, *dfrA1*+*aadA1*, *dfrA17*+*aadA5*, *dfrA12*, *dfrA1*+*sat2*+*aadA1*, *dfrA12*+*aadA2*, *dfrA5*, *aadA2*+*ere(A)*+*dfrA32*, *aac(6')*-*Ib-cr*, *aadA22*, *aadA5*, *dfrA17*, and *dfrA27*. Moreover, 13 MGEs in 69 different combinations were observed with predominance of IS26 followed by *tnpA*/Tn21, *trbC*, *ISEcp1*, *merA*, *ISAba1*, *tnsA*, *tnsB*, *tnsC*, *IS1133*, *tnsD*, *ISCR3/14*, and *tnsE*. Thus, the monitoring of integrons, MGEs and ARGs is important to understand the complex mechanism of AMR, which might help to introduce interventions for prevention and control of AMR in duck farms in China.

1. Introduction

Healthy ecosystem is required for public health security. The World Health Organization is considering “One Health” approach to tackle the global problem of antimicrobial resistance (AMR), that is directly depending on close association among animals, humans, and environmental health (Tiedje et al., 2019; Dafale et al., 2020). The AMR is a

rapidly growing health apprehension worldwide in both human and animal medicine. The extensive use of antimicrobial drugs is in practice for treating bacterial infections (Chen et al., 2014). The use or misuse of drugs has significantly contributed to the rise of AMR, which is alarming for therapeutics and prevention of bacterial diseases in veterinary and public health (Yassin et al., 2017; Dafale et al., 2020). Worryingly, even medical students need improvements in the knowledge of AMR, as a

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study in medical undergraduates was found many misconceptions about AMR (Assar et al., 2020). Bacteria are evolving themselves to fight against antimicrobials, by capturing or spreading antibiotic resistant genes (ARGs) and mobile genetic elements (MGEs) among the bacterial genes pool. Generalized MGEs may include partial genes, which exist in the transposon, phage, insertion sequence and plasmid (Partridge et al., 2018; Zhang et al., 2020). The transmission of ARGs/MGEs might happen by horizontal gene transfer (HGT) not only within bacterial species but also across different bacterial species (Brown-Jaque et al., 2015). MGEs are often linked with ARGs, leading AMR to spread easily among bacterial species (Karkman et al., 2018). Moreover, the gene cassettes (GCs) are gene fragments on variable regions (VRs) of integrons, which often carry specific ARGs. The GCs are usually transmitted by integrons mediated by other MGEs, eventually leading to the contamination of the ecological environment (Ravi et al., 2014; Zhang et al., 2020; Chaturvedi et al., 2021b), and our previous study have showed clinically healthy waterfowl birds could act as a reservoir for *Escherichia coli* (*E. coli*) with various GCs/ARGs, which might be a threat to public health and environmental safety (Zhang et al., 2019).

Many studies have explored the dissemination of resistance phenotype traits, genetic factors and AMR (Na et al., 2019; Bourély et al., 2019; Singh et al., 2021). However, limited published work was available on occurrences of integrons in *E. coli* from ducks in China (Zhang et al., 2019). The high occurrence of MGEs has recognized from large-scale poultry in China, which might participate in rising frequencies of AMR (Zhao et al., 2020). The involvement of MGEs in the transmission of AMR within humans and animals is essential (Zhang et al., 2020) and advance accessibility of gene information is vital in enhancing our knowledge of AMR transmission at the human-livestock interface (Wee et al., 2020).

Duck products favored by consumers are traditional good quality protein foods in China. As per “World Waterfowl Industry Development Conference, 2020”, a large number of consumable waterfowl birds are raised in China (Zhang et al., 2021a). Environmental safety is a major concern with increasing feeding and consumption of ducks, due to the widespread existence of multi-drug resistant (MDR) *E. coli* in this industry. The Sichuan province situated in the southwest of China, which is a landlocked basin with various types of wetland and diverse waterfowl species, and this area is also well-known for food and tourist point in China. The prevalence of integrons, ARGs, and GCs in *E. coli* isolated from food-producing waterfowl birds is under progress in Hainan, China (Zhang et al., 2019). In the other hand, avian pathogenic *Escherichia coli* (APEC) with harmfulness can cause pericarditis in waterfowls, which is often linked with several major serotypes, such as O1, O2, O18 and O78. (Ewers et al., 2004; Wang et al., 2014; Xu et al., 2019; Goudarzalejerdi et al., 2020). Thus, this study was designed to gather advance information about the diversity of MGEs, ARGs, integrons and resistance profiles in pathogenic and nonpathogenic *E. coli* isolates from the pericardial effusion or intestinal contents of diseased ducks. Notably, the sampling site depended on the clinical symptoms and signs. Moreover, this study was aimed to elaborate the occurrence of integrons, GCs, MGEs and resistance profiles, which represented the current situation of AMR and prevalence of MDR *E. coli* in the Sichuan province, China. Finally, this work highlights a need to impose intervention measures in the duck industry.

2. Material and methods

2.1. Isolation of bacterial strains

The *E. coli* strains were isolated from the pericardial effusion or intestinal contents of ducks that died of some diseases in duck farms around the Sichuan Agricultural University, Chengdu, China. Total 192 carcasses of diseased ducks were sent by farmers to the laboratory in cold temperature for testing, from the year of 2018–2020. Briefly, samples were taken from the pericardial effusion with clinical symptoms

of pericarditis, while others were taken from the intestinal contents. The samples were inoculated in LB broth and streaked on MacConkey agar (AOBOXING, Beijing, China, #02–005A). The pink colonies on MacConkey agar were further verified on CHROMagar (CHROMagar, Paris, France, #EC166), as well as identified by 16S rRNA sequencing with universal primers. The verified and confirmed *E. coli* isolates were re-suspended in tryptic soya broth plus 20% glycerol and stored at -80°C for further experiments.

2.2. Determination of Antimicrobial susceptibility

The antimicrobial susceptibilities were determined for all isolates against 12 antimicrobial agents via a standard disk diffusion test as recommended by (CLSI, 2019). The tested antimicrobial drugs disks (Oxoid Ltd., Basingstoke, United Kingdom) were amoxicillin/clavulanic acid 2:1 (AMC, 20/10 μg), ampicillin (AMP, 10 μg), ciprofloxacin (CIP, 5 μg), amikacin (AK, 30 μg), streptomycin (S, 10 μg), cefotaxime (CTX, 30 μg), ertapenem (ETP, 10 μg), doxycycline (DO, 30 μg), chloramphenicol (C, 30 μg), trimethoprim-sulfamethoxazole (SXT, 23.75/1.25 μg), polymyxin B (PB, 300 units) and rifampin (RD, 5 μg). *E. coli* ATCC25922 was used as a control strain. The antibiotics were selected for testing which was usually used in therapeutics, control and prevention of diseases in animal farms both in local and other places (Magiorakos et al., 2012; Gong et al., 2013; Bourély et al., 2019). The bacterial strains resistant to three and more classes of antimicrobials were categorized as MDR (Magiorakos et al., 2012).

2.3. DNA extraction

Total bacterial DNA was taken out from each isolate by heat extraction method as described (Yang et al., 2019) and stored at -80°C for further use. Briefly, 1 ml overnight incubated culture was taken in a sterile Eppendorf tube subjected to centrifugation at 13,000 rpm for 5 mins at 4°C . The pellet was taken in 300 μl of sterile water and placed in a boiling water bath (95°C) for 15 min and subjected to -20°C for 10 mins. The sample was then centrifuged at 13,000 g for 5 min and the supernatant containing DNA was collected. Finally, the DNA qualities were checked via ultraviolet-absorbance (ND1000, Nanodrop, Thermo Fisher Scientific).

2.4. Screening for O serotype testing, integrons, GCs, MGEs and sequence data analysis

O serotypes (O1, O2, O18 and O78) linked with APEC were screened via primers as described by (Li et al., 2010; Liu et al., 2010; Wang et al., 2014), and the *E. coli* isolates were screened for *int1*, *int2* and *int3* integrase genes by polymerase chain reaction (PCR) as per previously described methodology (Table S1).

The integrase gene-positive strains were subjected to PCR for the elaboration of VRs. The amplification and screening of GCs in VRs was performed. Further analyses, primers specific to the VRs of class 1 and 2 integrons were used for the amplification of the integrons-GCs array, respectively. In addition, 3'-conserved segment (*qacEA1*+*sul1*) of *int1* positive *E. coli* and MGEs genes were also screened by PCR (Table S1). All positive PCR products were sequenced by Beijing Genomics Institute (BGI), CO., Ltd. (Chengdu, China). Finally, the sequences were compared and analysed via the online gene bank BLAST tool (<http://www.ncbi.nlm.nih.gov/BLAST/>), followed by submitting sequences of GCs array to the public database of NCBI.

2.5. Statistical analysis

Bacterial isolates were categorized as susceptible, resistant, and intermediate/moderately resistant based on the zone of inhibition diameter interpretative as per the standard mentioned by (CLSI, 2019). The revealed data were statistically analysed by Fisher's exact test using

SPSS Statistics version 21.0. A Fisher's exact test was applied to associate the AMR phenotype to integrons in isolates obtained from diseased ducks in the Sichuan province of China. Differences were measured significant at $P < 0.05$.

3. Results

3.1. Isolation of *E. coli* from diseased ducks

To examine carcasses of diseased duck, one isolate per pericardial effusion or intestinal content was taken in the clean and sterilized animal room, which was mainly based on clinical symptoms of ducks. A total of 165 non-repetitive *E. coli* were isolated from 2018 to 2020 in the present experiment. Largely, the isolation rate of samples was 85.9%. Among them, 16 *E. coli* strains were linked with O1, O2 and O78 serotypes, i.e. PZ18, DJY64, PD67, PZ73, PZ75, PD93 ($n = 6$, O1), DJY52, LS114 ($n = 2$, O2), QL127–131 and QL139–141 ($n = 8$, O78). Moreover, 165 bacterial strains were revealed as PZ1–PZ50, DJY51–DJY65, PD66–PD68, PZ69–PZ78, PD79–PD94, dy95–dy108, LS109–LS121, QL122–QL132, MS133–MS135, DY136–DY137, QL138–QL151, CZ152–CZ154, DY155–DY158, MS159, DY160–DY161, and CZ162–CZ165. The initial letters of strain names PZ, DJY, PD, dy, LS, QL, MS, DY, and CZ were representing geographical regions of isolates named Pengzhou, Duijiangyan, Pidun, Dayi, Leshan, Qionglai, Meishan, Deyang and Chongzhou, respectively, which were all located in Sichuan Province of China.

3.2. Antimicrobial susceptibility analysis

The AMR profile of the examined isolates was given (Table 1 and Figs 1 and S1). Most of the *E. coli* strains isolated from diseased ducks were resistant to more than three classes of antimicrobials and the majority of MDR strains were detected in integron positive isolates (Fig. 2). Compared with integron-negative strains, integron-positive isolates had higher drug resistance. Among them, there were 1, 8, 43, 88 and 4 integron positive isolates resistant to 1–2, 3–4, 5–6, 7–8 and 9–10 antibiotics, respectively, while integron negative isolates were only 1, 1, 10, 9 and 0 (Fig. 2). Briefly, there was 100% ($n = 165$) isolates resistant to at least one antibiotic and 98.8% ($n = 163$) were MDR strains. In particular, PD68 was only resistant to RD, and dy98 was resistant to SXT and RD. Besides, Only DY156 showed resistant to ETP. Outcomes from 165 isolates exhibited highest resistance to RD (97.0%) followed by AMP (95.8%), C (89.7%), SXT (84.2%), CIP (83.0%), CTX (80.0%), S

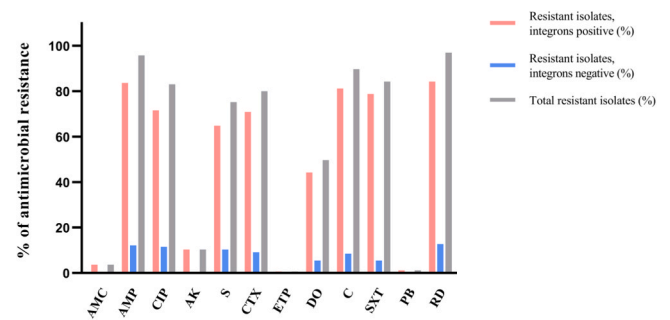


Fig. 1. Antimicrobial resistant to 12 antibiotics showed by *E. coli* isolates which were isolated from the intestinal contents or pericardial effusion of ducks that died of diseases in Sichuan, China. The columns show the resistant percentages against antibiotics amoxicillin/clavulanic acid (AMC), ampicillin (AMP), ciprofloxacin (CIP), amikacin (AK), streptomycin (S), cefotaxime (CTX), eropenem (ETP), doxycycline (DO), chloramphenicol (C), trimethoprim-sulfamethoxazole (SXT), polymyxin B (PB), and rifampin (RD).

(75.8%), DO (49.7%), AK (10.3%), AMC (3.6%), PB (1.2%) and ETP (0.6%). Results were interpreted following CLSI (2019) criteria.

3.3. Occurrence of Integrons and gene cassettes

Out of the 165 *E. coli* isolates obtained from intestinal contents or pericardial effusion of diseased ducks, 87.3% ($n = 144$) and 17.6% ($n = 29$) isolates were found positive for *int1* and *int2* genes, respectively. The 17.6% ($n = 29$) isolates were revealed positive for both *int1* and *int2* genes. However, no isolate was positive for *int3* gene, and 12.7% ($n = 21$) isolates were negative for any *int* gene. Further, VRs of *int1* gene positive isolates were also explored in the present study. To be exact, total 62 GCs (58 GCs in class 1 integron, 4 GCs in class 2 integron) were detected in 61 *E. coli* isolates, among which PZ50 contained both class 1 and 2 integrons-GCs. Moreover, only one kind of GC (*dfrA1+sat2+aadA1*) was detected in VRs of class 2 integron (Table 2). In particular, *E. coli* isolates with class 1 and 2 integron showed 13 kinds of GCs via screening of sequences, with predominance of *arr-3+dfrA27* (9.1%, $n = 15$). The other prevalent GCs were including *dfrA1+aadA1* (6.7%, $n = 11$), *dfrA17+aadA5* (6.1%, $n = 10$), *dfrA12* (3.0%, $n = 5$), *dfrA1+sat2+aadA1* (2.4%, $n = 4$), *dfrA12+aadA2* (2.4%, $n = 4$), *dfrA5* (2.4%, $n = 4$), *aadA2+ere(A)+dfrA32* (1.8%, $n = 3$), and *aac(6')-Ib-cr* (1.2%, $n = 2$). Further, *aadA22*, *aadA5*, *dfrA17*, and *dfrA27* genes were found in single *E. coli* isolate LS111, dy108, DJY55, and PZ42,

Table 1

Association between the occurrence of integrons and AMR phenotype in *E. coli* isolates from diseased ducks in Sichuan ($n = 165$).

Antibiotic	No. of resistant isolates, integrons positive (%) ^a	No. of resistant isolates, integrons negative (%) ^b	Total no. of resistant isolates (%)	No. of sensitive isolates, integrons positive (%)	No. of sensitive isolates, integrons negative (%)	No. of moderately resistant isolates, integrons positive (%)	No. of moderately resistant isolates, integrons negative (%)	P-value ^c
AMC	6 (3.6)	0 (0.0)	6 (3.6)	114 (69.1)	16 (9.7)	24 (14.5)	5 (3.0)	1.000
AMP	138 (83.6)	20 (12.1)	158 (95.8)	4 (2.4)	1 (0.6)	2 (1.2)	0 (0.0)	1.000
CIP	118 (71.5)	19 (11.5)	137 (83.0)	16 (9.7)	2 (1.2)	10 (6.1)	0 (0.0)	0.534
AK	17 (10.3)	0 (0.0)	17 (10.3)	127 (77.0)	21 (12.7)	0 (0.0)	0 (0.0)	0.132
S	108 (65.5)	17 (10.3)	125 (75.8)	22 (13.3)	4 (2.4)	14 (8.5)	0 (0.0)	0.786
CTX	117 (70.9)	15 (9.1)	132 (80.0)	25 (15.2)	6 (3.6)	2 (1.2)	0 (0.0)	0.379
ETP	1 (0.6)	0 (0.0)	1 (0.6)	143 (86.7)	21 (12.7)	0 (0.0)	0 (0.0)	1.000
DO	73 (44.2)	9 (5.5)	82 (49.7)	27 (16.4)	6 (3.6)	44 (26.7)	6 (3.6)	0.641
C	134 (81.2)	14 (8.5)	148 (89.7)	2 (1.2)	2 (1.2)	8 (4.8)	5 (3.0)	0.002
SXT	130 (78.8)	9 (5.5)	139 (84.2)	12 (7.3)	12 (7.3)	2 (1.2)	0 (0.0)	0.000
PB	2 (1.2)	0 (0.0)	2 (1.2)	141 (85.5)	21 (12.7)	1 (0.6)	0 (0.0)	1.000
RD	139 (84.2)	21 (12.7)	160 (97.0)	0 (0.0)	0 (0.0)	5 (3.0)	0 (0.0)	1.000

Note: AMC: amoxicillin/clavulanic acid; AMP: ampicillin; CIP: ciprofloxacin; AK: amikacin; S: streptomycin; CTX: cefotaxim; ETP: eropenem; DO: doxycycline; C: chloramphenicol; SXT: trimethoprim-sulfamethoxazole; PB: polymyxin B; RD: rifampin.

^a No. of AMR phenotype isolates, which are positive for integrase genes via PCR and sequencing.

^b No. of AMR phenotype isolates, which are negative for integrase genes via PCR and sequencing.

^c Significant values are in bold.

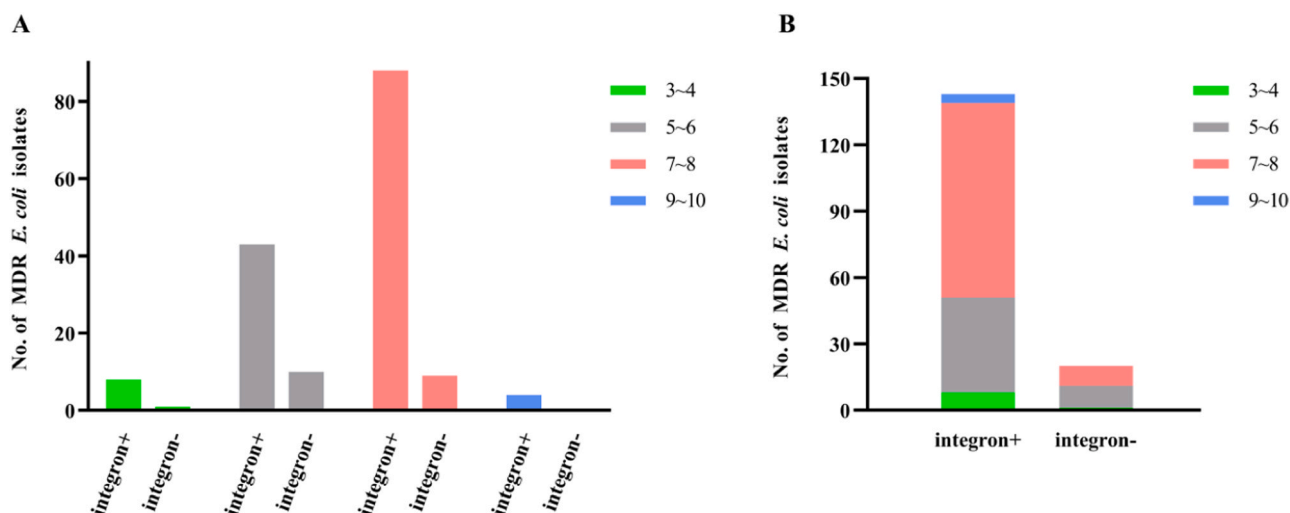


Fig. 2. High rate of AMR patterns shown by *E. coli* isolates from diseased ducks, and number of different classes of antibiotics resistance was listed. (A) The antibiotic resistance revealed by *E. coli* isolates against up to 3–4 antimicrobials were shown in green color, to 5–6 antimicrobials in gray color, to 7–8 antimicrobials in pink color and to 9–10 antimicrobials in blue color. Most isolates were resistant to 7–8 antimicrobials followed by 5–6 antimicrobials. (B) MDR was exhibited by number of strains with positive and negative integrons. The worst MDR was observed in integron positive isolates. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

respectively. However, there were no GCs found in 86 or 25 isolates, which contained class 1 or 2 integrons, respectively. The sequences of all GCs array identified in the present study were submitted to NCBI GenBank. The genbank accession numbers, detailed resistance profile and GCs of isolates were presented in Table 2.

3.4. Relationship among AMR phenotype and integrons

The association between AMR phenotypes and the occurrence of integrons was elaborated in this study, then the different resistance phenotypes to SXT and C were significantly associated with the existence of integrons (Table 1). In addition, no significant association was found in other AMR phenotypes to the presence of integrons, including AMC, AMP, CIP, AK, S, CTX, ETP, DO, PB and RD. However, almost all the *E. coli* isolates exhibited corresponding AMR profiles to the GCs harbored in their integrons (Table 2).

3.5. Detection of resistance to sulfonamides, and quaternary amines

The *qacEA1* and *sul1* genes in class 1 integron were analysed in 165 isolates via PCR. The *qacEA1-sul1* genes were part of 3'-conserved segment of class 1 integron which were conferring resistance to sulfonamide and quaternary ammonium compounds, respectively. The *sul1* gene was observed in (60.6%, $n = 100$), and *qacEA1* gene was present in (58.8%, $n = 97$). Further, the isolates containing *sul1+*, *qacEA1+* (53.3%, $n = 88$); *sul1-*, *qacEA1-* (33.9%, $n = 56$); *sul1+*, *qacEA1-* (7.3%, $n = 12$) and *sul1-*, *qacEA1+* (5.5%, $n = 9$) were recorded.

3.6. Analysis and distribution of mobile elements

In the present study, 13 MGEs were obtained in 165 isolates with the predominance of IS26 (100%, $n = 165$). Other MGEs recorded were including *tnpA*/Tn21 (64.8%, $n = 107$), *trbC* (58.2%, $n = 96$), *ISEcp1* (38.2%, $n = 63$), *merA* (26.1%, $n = 43$), *ISAbal* (21.8%, $n = 36$), *tnsA* (8.5%, $n = 14$), *tnsB* (8.5%, $n = 14$), *tnsC* (8.5%, $n = 14$), *IS1133* (6.7%, $n = 11$), *tnsD* (6.1%, $n = 10$), *ISCR3/14* (3.6%, $n = 6$), *tnsE* (3.6%, $n = 6$). All isolates were negative for IS903. The four most common combinations of MGEs were including *int11*+IS26+*trbC* (7.9%, $n = 13$) followed by *int11*+IS26+*tnpA*/Tn21 (6.7%, $n = 11$), *int11*+IS26 (6.1%, $n = 10$), and *int11*+IS26+*trbC*+*ISEcp1*+*tnpA*/Tn21 (5.5%, $n = 9$). The MGEs were found in total 69 different combinations in these *E. coli*

isolates (Table 3). To the best of our knowledge, this is a detailed study reporting high occurrences and combinations of MGEs in *E. coli* isolated from intestinal contents or pericardial effusion of diseased ducks from Sichuan province, China.

4. Discussion

Due to the prevalence of waterfowl bacterial infectious diseases and the extensive use of antibiotics, the emergence of MDR bacteria with integrons and MGEs has been accelerated in this industry, which in turn led to more irrational use of antibiotics, subsequently leading to the widespread of ARGs and drug residues in meat, eggs and other animal derived food. This vicious circle has seriously threatened human health and environmental safety. Therefore, waterfowl health should also be considered to be related to human health, which was forming a unique perspective for the concept of "One Health".

In the present study, we have detected 98.8% ($n = 163$) phenotypically MDR *E. coli* obtained from the intestinal contents/pericardial effusion of diseased ducks carcasses. We found high resistance to frequently used antimicrobials in more than 80.0% of *E. coli*, such as RD, AMP, C, SXT, CIP and CTX. Originating from clinically healthy waterfowls, we observed relatively lower level of MDR (80.7%) in *E. coli* and resistance phenotypes to C (64.3%), SXT (67.2%), CIP (34.1%), and CTX (49.2%) (Zhang et al., 2021a). However, significantly rising AMR trends were observed in 17 provinces of China from 1993 to 2013 to antimicrobial agents including AK, AMP, CTX, C, CIP, and SXT, which was similar to findings of this study (Chen et al., 2014). In current research, slightly lower MDR was observed in comparison to previously reported 100% MDR from China (Yassin et al., 2017), but higher than the finding in South Korea, where 54% of duck isolates were resistant to three or more antimicrobial drugs (Na et al., 2019). Further, in South Korea, the resistance rate was noted at a lower level to SXT (51.3%), AMP (47.8%), and CIP (23.0%), whereas resistance to AMC was not detected (Na et al., 2019). Moreover, the French surveillance network for antimicrobial resistance in diseased animals analysed isolates from 2012 to 2016 in seven animals species, where duck isolates were shown 39.7% MDR (Bourély et al., 2019), much lower than our results. The variations in MDR among countries might be due to various degree of antibiotics usage, geographical regional differences, and transmission capacity of resistance.

Though at a lower rate but resistance to ETP and AMC was observed

Table 2Resistance profile and integrons-GCs in *E. coli* isolates from diseased ducks in Sichuan.

Strains	Resistance profile	Integron	Integrons-GCs ^a	Accession number	O serotype	Source ^c
PZ14	AMP/CIP/AK/CTX/DO/C/SXT/RD	class1+class2	<i>aac(6')-Ib-cr</i>	MW430670	d	I
PZ22	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW430671	d	I
PZ23	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW430672	d	I
PZ24	AMP/CIP/S/CTX/DO/C/SXT/RD	class1+class2	<i>arr-3+dfra27</i>	MW326770	d	I
PZ25	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW521221	d	I
PZ32	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW521222	d	I
PZ34	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW521223	d	I
PZ35	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW521224	d	I
PZ36	AMP/CIP/DO/C/SXT/RD	class1	<i>dfra5</i>	MW521225	d	I
PZ37	AMP/CTX/DO/C/SXT/RD	class1+class2	<i>dfra5</i>	MW521226	d	I
PZ38	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>dfra5</i>	MW521227	d	I
PZ41	AMP/CIP/S/SXT/RD	class1	<i>dfra17+aadA5</i>	MW521228	d	I
PZ42	AMP/S/CTX/DO/C/SXT/RD	class1	<i>dfra27</i>	MW521229	d	I
PZ44	AMP/CIP/CTX/C/SXT/RD	class1	<i>dfra17+aadA5</i>	MW521230	d	I
PZ45	AMP/CIP/C/SXT/RD	class1+class2	<i>dfra17+aadA5</i>	MW521231	d	I
PZ49	AMP/CIP/S/CTX/DO/C/SXT/PB/RD	class1	<i>dfra17+aadA5</i>	MW521232	d	I
PZ50	AMP/CIP/S/CTX/C/SXT/RD	class1+class2	<i>dfra5/dfra1+sat2+aadA1^b</i>	MW535270/MW521233	d	I
DJY52	AMP/CIP/S/CTX/DO/C/SXT/RD	class1+class2	<i>dfra1+sat2+aadA1^b</i>	MW521234	O2	I
DJY53	AMP/CIP/S/CTX/DO/C/SXT/RD	class1+class2	<i>dfra1+sat2+aadA1^b</i>	MW521235	d	I
DJY54	AMP/CIP/S/CTX/DO/C/SXT/RD	class1+class2	<i>dfra1+sat2+aadA1^b</i>	MW521236	d	I
DJY55	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>dfra17</i>	MW521237	d	I
PZ69	AMP/CIP/CTX/C/SXT/RD	class1	<i>dfra12+aadA2</i>	MW521238	d	I
PZ71	AMP/CIP/CTX/C/SXT/RD	class1	<i>dfra12+aadA2</i>	MW521239	d	I
PD92	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW521240	d	I
dy99	AMP/CIP/S/DO/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW430666	d	I
dy108	AMP/AK/S/CTX/C/SXT/RD	class1	<i>aadA5</i>	MW430667	d	I
LS111	AMP/S/CTX/DO/C/RD	class1	<i>aadA22</i>	MW430668	d	I
LS119	AMP/CIP/AK/S/CTX/DO/C/SXT/RD	class1	<i>dfra12+aadA2</i>	MW430669	d	I
MS133	AMP/CIP/AK/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW535271	d	P
MS134	AMP/CIP/S/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW535272	d	I
MS135	AMP/CIP/S/SXT/RD	class1	<i>dfra1+aadA1</i>	MW535273	d	I
DY136	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW535274	d	I
DY137	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW535275	d	I
QL138	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra17+aadA5</i>	MW535276	d	P
QL139	AMP/CIP/AK/S/CTX/C/SXT/RD	class1	<i>aadA2+ere(A)+dfra32</i>	MW535277	O78	P
QL140	AMP/CIP/AK/S/CTX/C/SXT/RD	class1	<i>aadA2+ere(A)+dfra32</i>	MW535278	O78	P
QL141	AMP/CIP/AK/S/CTX/C/SXT/RD	class1+class2	<i>aadA2+ere(A)+dfra32</i>	MW535279	O78	P
QL142	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW535280	d	P
QL143	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW535281	d	P
QL144	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra17+aadA5</i>	MW535282	d	P
QL145	AMP/CIP/S/C/SXT/RD	class1	<i>dfra12+aadA2</i>	MW535283	d	P
QL146	AMP/CIP/S/C/SXT/RD	class1	<i>dfra12</i>	MW535284	d	I
QL147	AMP/CIP/S/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW535285	d	I
QL148	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW535286	d	I
QL149	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW535287	d	I
QL150	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra17+aadA5</i>	MW535288	d	I
QL151	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra12</i>	MW546921	d	I
CZ152	C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW546922	d	I
CZ153	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra1+aadA1</i>	MW546923	d	I
CZ154	AMC/AMP/CIP/S/CTX/DO/C/SXT/PB/RD	class1	<i>aac(6')-Ib-cr</i>	MW579371	d	I
DY155	AMP/CIP/S/CTX/C/SXT/RD	class1+class2	<i>dfra1+aadA1</i>	MW546920	d	I
DY156	AMC/AMP/S/CTX/ETP/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW546924	d	I
DY157	AMP/CIP/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW546925	d	I
DY158	AMP/CIP/CTX/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW546926	d	I
MS159	AMP/CIP/AK/CTX/DO/C/SXT/RD	class1	<i>arr-3+dfra27</i>	MW546927	d	I
DY160	AMP/CIP/S/CTX/DO/C/SXT/RD	class1+class2	<i>dfra17+aadA5</i>	MW546928	d	I
DY161	AMP/CIP/S/CTX/C/SXT/RD	class1+class2	<i>dfra12</i>	MW546929	d	I
CZ162	AMP/CIP/S/C/SXT/RD	class1	<i>dfra12</i>	MW546930	d	I
CZ163	AMP/CIP/S/CTX/C/SXT/RD	class1	<i>dfra12</i>	MW546931	d	I
CZ164	AMP/CIP/CTX/C/SXT/RD	class1	<i>dfra17+aadA5</i>	MW546932	d	I
CZ165	AMP/S/CTX/DO/C/SXT/RD	class1	<i>dfra17+aadA5</i>	MW546933	d	I

Note: **AMC**: amoxicillin/clavulanic acid; **AMP**: ampicillin; **CIP**: ciprofloxacin; **AK**: amikacin; **S**: streptomycin; **CTX**: cefotaxim; **ETP**: ertapenem; **DO**: doxycycline; **C**: chloramphenicol; **SXT**: trimethoprim-sulfamethoxazole; **PB**: polymyxin B; **RD**: rifampin.

^a *dfra1*, *dfra5*, *dfra12*, *dfra17*, *dfra27* and *dfra32* were conferring resistance to trimethoprim; *aadA1*, *aadA2*, *aadA5* and *aadA22* were conferring resistance to streptomycin and spectinomycin; *sat2* was conferring resistance to streptomycin; *ere(A)* was conferring resistance to erythromycin; *aac(6')-Ib-cr* was conferring resistance to aminoglycosides and fluoroquinolones; *arr-3* was conferring resistance to rifampin.

^b *dfra1+sat2+aadA1* was found in GC array of class 2 integron, and no other GCs were detected in class 2 integron.

^c "I" stands for intestinal contents, "P" stands for pericardial effusion.

^d indicates that no related serotype was detected.

Table 3Components of MGEs in *E. coli* isolates from diseased ducks in Sichuan ($n = 165$).

Components of MGEs	Number (%)	Isolates
<i>int1</i> +IS26+trbC	13 (7.9)	PZ22, PZ25, PZ32, PZ77, PD81, dy100, dy107, LS109, LS110, LS117, LS121, QL146, QL149
<i>int1</i> +IS26+tnpA/Tn21	11 (6.7)	PZ23, PD94, dy102, LS113, QL125, QL127, QL128, QL129, QL131, QL132, DY156
<i>int1</i> +IS26	10 (6.1)	PZ34, PZ35, PZ36, dy95, dy96, dy97, dy98, dy99, LS112, QL123
<i>int1</i> +IS26+trbC+ISEcp1+tnpA/Tn21	9 (5.5)	PZ1, PZ2, PZ3, PZ20, PZ27, PZ31, PZ47, PZ69, PZ75
<i>int1</i> +IS26+trbC+tnpA/Tn21	8 (4.8)	PZ72, PD82, dy108, QL130, QL143, CZ162, CZ163, CZ164
<i>int1</i> +merA+IS26	6 (3.6)	PZ6, PZ7, PZ8, PZ10, PZ11, PZ78
<i>int1</i> +IS26+ISEcp1+tnpA/Tn21	5 (3.0)	PZ28, PZ33, PZ46, PZ48, QL144
IS26+trbC	5 (3.0)	PZ4, PD79, PD80, dy104, LS116
<i>int1</i> +IS26+ISEcp1	4 (2.4)	PZ39, dy101, LS119, QL138
<i>int1</i> +ISAbal+IS26+ISEcp1+tnpA/Tn21	4 (2.4)	PZ17, LS114, DY158, MS159
<i>int1</i> +ISAbal+IS26+tnpA/Tn21	4 (2.4)	PD92, dy106, CZ152, DY157
<i>int1</i> +merA+IS26+trbC+tnpA/Tn21	4 (2.4)	PZ13, DJY65, QL124, MS135
<i>int1</i> +merA+IS26+tnpA/Tn21	5 (3.0)	PZ12, PZ30, LS111, QL126, DY137
IS26+trbC+tnpA/Tn21	4 (2.4)	PD83, PD84, PD85, PD87
<i>int1</i> + <i>int1</i> 2+IS26+trbC+ISEcp1+tnpA/Tn21	3 (1.8)	PZ5, PZ19, DJY62
<i>int1</i> +IS26+trbC+ISEcp1	3 (1.8)	PZ21, LS118, CZ165
IS26	3 (1.8)	PD68, dy105, QL122
<i>int1</i> + <i>int1</i> 2+IS26+trbC+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC	2 (1.2)	PZ29, PZ74
<i>int1</i> + <i>int1</i> 2+IS26+trbC+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC+tnsD+tnsE	2 (1.2)	DJY60, DY160
<i>int1</i> + <i>int1</i> 2+merA+IS26	2 (1.2)	PZ9, DY155
<i>int1</i> + <i>int1</i> 2+merA+IS26+trbC+ISEcp1+tnpA/Tn21	2 (1.2)	PZ14, DJY63
<i>int1</i> +ISAbal+IS26+trbC+ISEcp1+tnpA/Tn21	2 (1.2)	PZ26, LS115
<i>int1</i> +ISAbal+IS26+trbC+tnpA/Tn21	2 (1.2)	PZ70, PD93
<i>int1</i> +merA+IS26+trbC+ISEcp1	3 (1.8)	PZ38, PZ41, PZ71
<i>int1</i> +merA+IS26+trbC+ISEcp1+tnpA/Tn21	2 (1.2)	PZ15, DJY57
<i>int1</i> +merA+ISAbal+IS26+trbC+IS1133+tnpA/Tn21	2 (1.2)	QL147, QL148
<i>int1</i> +merA+ISAbal+IS26+trbC+ISEcp1+tnpA/Tn21	2 (1.2)	PZ18, MS133
IS26+tnpA/Tn21	2 (1.2)	PD86, PD91
<i>int1</i> + <i>int1</i> 2+IS26+ISEcp1+tnpA/Tn21	1 (0.6)	PD67
<i>int1</i> + <i>int1</i> 2+IS26+trbC+IS1133+tnpA/Tn21	1 (0.6)	PZ45
<i>int1</i> + <i>int1</i> 2+IS26+trbC+ISEcp1	1 (0.6)	PD66
<i>int1</i> + <i>int1</i> 2+ISAbal+IS26+trbC+ISEcp1+tnpA/Tn21	1 (0.6)	PZ16
<i>int1</i> + <i>int1</i> 2+ISAbal+IS26+trbC+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC	1 (0.6)	DJY61
<i>int1</i> + <i>int1</i> 2+ISAbal+IS26+trbC+tnpA/Tn21+tnsA+tnsB+tnsC	1 (0.6)	PZ73
<i>int1</i> + <i>int1</i> 2+ISAbal+IS26+trbC+tnpA/Tn21+tnsA+tnsB+tnsC+tnsD	1 (0.6)	DJY54
<i>int1</i> + <i>int1</i> 2+ISAbal+IS26+tnpA/Tn21	1 (0.6)	DY161
<i>int1</i> + <i>int1</i> 2+merA+IS26+trbC+ISEcp1	1 (0.6)	PZ24
<i>int1</i> + <i>int1</i> 2+merA+IS26+trbC+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC	1 (0.6)	DJY64
<i>int1</i> + <i>int1</i> 2+merA+IS26+trbC+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC+tnsE	1 (0.6)	DJY51
<i>int1</i> + <i>int1</i> 2+merA+IS26+trbC+tnpA/Tn21+tnsD	1 (0.6)	DJY56
<i>int1</i> + <i>int1</i> 2+merA+ISAbal+IS26+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC	1 (0.6)	DJY59
<i>int1</i> + <i>int1</i> 2+merA+ISAbal+IS26+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC+tnsD	1 (0.6)	PZ50
<i>int1</i> + <i>int1</i> 2+merA+ISAbal+IS26+trbC	1 (0.6)	PZ37
<i>int1</i> + <i>int1</i> 2+merA+ISAbal+IS26+trbC+IS1133+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC+tnsD+tnsE	1 (0.6)	DJY52
<i>int1</i> + <i>int1</i> 2+merA+ISAbal+IS26+trbC+ISEcp1+tnpA/Tn21+tnsA+tnsB+tnsC+tnsD+tnsE	1 (0.6)	DJY53
<i>int1</i> +IS26+trbC+tnpA/Tn21	1 (0.6)	PZ42
<i>int1</i> +IS26+IS1133	1 (0.6)	LS120
<i>int1</i> +IS26+IS1133+ISEcp1+tnpA/Tn21	1 (0.6)	QL150
<i>int1</i> +IS26+IS1133+tnpA/Tn21	1 (0.6)	PZ44
<i>int1</i> +IS26+trbC+IS1133+ISEcp1+tnpA/Tn21	1 (0.6)	PZ43
<i>int1</i> +IS26+trbC+IS1133+tnpA/Tn21	1 (0.6)	QL151
<i>int1</i> +IS26+trbC+ISCR3/14	1 (0.6)	QL145
<i>int1</i> +IS26+trbC+tnpA/Tn21+ISCR3/14	1 (0.6)	QL142
<i>int1</i> +ISAbal+IS26+IS1133+ISEcp1+tnpA/Tn21	1 (0.6)	PZ49

(continued on next page)

Table 3 (continued)

Components of MGEs	Number (%)	Isolates
<i>intI1</i> + <i>merA</i> + <i>IS26</i> + <i>trbC</i>	1 (0.6)	DY136
<i>intI1</i> + <i>merA</i> + <i>IS26</i> + <i>trbC</i> + <i>tnpA</i> /Tn21+ <i>ISCR3/14</i>	1 (0.6)	MS134
<i>intI1</i> + <i>merA</i> + <i>ISAbal</i> + <i>IS26</i> + <i>trbC</i> + <i>ISEcp1</i> + <i>tnpA</i> /Tn21+ <i>tnsB</i> + <i>tnsC</i>	1 (0.6)	CZ154
<i>intI1</i> + <i>merA</i> + <i>ISAbal</i> + <i>IS26</i> + <i>trbC</i> + <i>ISEcp1</i> + <i>tnpA</i> /Tn21+ <i>tnsD</i> + <i>tnsE</i>	1 (0.6)	DJY55
<i>intI1</i> + <i>merA</i> + <i>ISAbal</i> + <i>IS26</i> + <i>tnpA</i> /Tn21	1 (0.6)	CZ153
<i>intI1</i> + <i>intI2</i> + <i>ISAbal</i> + <i>IS26</i> + <i>trbC</i> + <i>tnpA</i> /Tn21+ <i>ISCR3/14</i>	1 (0.6)	QL141
<i>IS26</i> + <i>trbC</i> + <i>ISEcp1</i>	1 (0.6)	dy103
<i>IS26</i> + <i>trbC</i> + <i>ISEcp1</i> + <i>tnpA</i> /Tn21	1 (0.6)	PD88
<i>IS26</i> + <i>trbC</i> + <i>ISEcp1</i> + <i>tnsD</i>	1 (0.6)	DJY58
<i>ISAbal</i> + <i>IS26</i> + <i>ISEcp1</i> + <i>tnpA</i> /Tn21	1 (0.6)	PD89
<i>intI1</i> + <i>ISAbal</i> + <i>IS26</i> + <i>trbC</i> + <i>IS1133</i> + <i>tnpA</i> /Tn21+ <i>ISCR3/14</i>	1 (0.6)	QL139
<i>intI1</i> + <i>ISAbal</i> + <i>IS26</i> + <i>trbC</i> + <i>tnpA</i> /Tn21+ <i>ISCR3/14</i>	1 (0.6)	QL140
<i>ISAbal</i> + <i>IS26</i> + <i>tnpA</i> /Tn21	1 (0.6)	PD90
<i>ISAbal</i> + <i>IS26</i> + <i>tnpA</i> /Tn21+ <i>tnsD</i>	1 (0.6)	PZ76
<i>merA</i> + <i>IS26</i>	1 (0.6)	PZ40

in our findings, which were previously reported zero in duck isolates (Yassin et al., 2017). Contradictory, from isolates of market chicken carcasses, resistance to AMC was observed at a considerably higher rate 48.5% (Meng et al., 2016). The resistance against β -lactams like AMP and CTX was reported 100% in Thailand, which was higher than our findings. Further, in this study, a rising trend of resistance rate to non- β -lactams such as SXT, C, CIP, AK (except DO) was observed, as compared to the previously reported resistance rate of these drugs i.e. 64.5%, 60.3%, 28.9% and 6.6% in Thailand, respectively (Tansawai et al., 2019). Recently, high resistance to AK (23.1%) was observed in fecal isolates of pigs (Li et al., 2020b). Similar to our findings, the higher levels of sulfonamides resistance have also been described in China, which might be due to the wide use of sulfonamides for a long time (Yassin et al., 2017). In the present study, 89.7% chloramphenicol resistance was recorded, which was significantly associated with the occurrence of integrons. Though, in 2002, the chloramphenicol has banned in food-producing animals in China (Ministry of Agriculture of the People's Republic of China MAPRC, 2002), subsequently replaced with florfenicol, which was approved to use in veterinary medicine by the Food and Drug Administration in 1996 (White et al., 2000). Therefore, it might be suggested that AMR sustain for a longer time and not easy to eliminate, once prompted.

Integrons, as the genetic elements, are known to capture one or more ARGs, which might be transferable among bacteria and also disseminate ARGs via transmissible plasmids, phage and insertion sequences, posing threat to public health (Zhang et al., 2020). High prevalence of class 1 (87.3%) and class 2 (17.6%) integrons was observed in this study, which showed a rising trend compared with our previous research of waterfowls in Hainan, China (Zhang et al., 2019). Total 13 types of GCs were revealed in class 1 or class 2 integrons-positive isolates (Table 2). The predominance of different GCs array variants of *dfrA* (*dfrA1*, *dfrA5*, *dfrA12*, *dfrA17*, *dfrA27* and *dfrA32*) and *aadA* (*aadA1*, *aadA2*, *aadA5* and *aadA22*) conferring resistance to trimethoprim and aminoglycosides, which was similar to previous reports in human (White et al., 2001; El-Najjar et al., 2010; Singh et al., 2021), in giant panda (Zou et al., 2018) and in yaks (Yang et al., 2017). The high prevalence of classes 1 and 2 integrons was found to affect water resources and environmental settings. For instance, integrons (*intI1* and *intI2*) were associated with *sul1* and *sul2* genes in riverine systems (Chaturvedi et al., 2021b). In addition, *intI1* and *intI2* genes were also found in 89.25% and 1.07% in Indian riverine isolates, respectively (Chaturvedi et al., 2021a). Further, *intI1* and *sul1* were found to at high levels in coastal marine water after the impact of hurricane events (Saingam et al., 2021). Worryingly, migratory water birds such as gulls, storks and cranes could disseminate high concentrations of *intI1* gene in environmental settings (Jarma et al., 2021). In the present study, both class 1 and 2 integrons positive isolates ($n = 4$) were containing GC (*dfrA1*+*sat2*+*aadA1*), which was recently reported from class 2 integrons in children (Singh et al., 2021), yaks (Yang et al., 2017), waterfowls (Zhang et al., 2019), and giant panda

(Zou et al., 2018). Moreover, the significant associations were found between SXT/C resistance phenotypes and the existence of integrons (Table 1), which was in line with a previous study of *Salmonella* isolates, with a higher prevalence of GC *dfrA17*+*aadA5* (59.6%) (Meng et al., 2016). Our findings showed the percentage of *E. coli* with integrons resistant to SXT were almost in line with the study of isolates from diseased chickens and swine in China (Yang et al., 2004).

It is to mention that O1, O2, and O78 *E. coli* (9.7%, $n = 16$) were isolated from dead ducks, which suggested that they might be closely related to APEC strains. These isolates separated from pericardial effusion were containing GCs *dfrA17*+*aadA5*, *aadA2*+*ere(A)*+*dfrA32*, *dfrA1*+*aadA1*, *dfrA12*+*aadA2*, and *arr-3*+*dfrA27*. A quite similar GCs conferring resistance to trimethoprim and aminoglycosides were reported from APEC in chickens (Awad et al., 2016). In the present study, an uncommon GC array *aadA2*+*ere(A)*+*dfrA32* ($n = 3$) was observed in O78 isolates, which was similar to the GC in *Proteus mirabilis* isolates from cooked meat (Jiang et al., 2017). Interestingly, all three isolates containing *aadA2*+*ere(A)*+*dfrA32* were negative for *sul1* and *qacEΔ1*. As far as we know, this unique gene cassette array was first found in *E. coli* of diseased ducks from Sichuan, China.

In addition to the above stated ARGs, another GC *aac(6')-Ib-cr* ($n = 2$) conferring resistance to aminoglycosides and fluoroquinolones was observed. Such GC was recently observed in urine isolates of clinical staff members (Li et al., 2020a). Further, absolute abundance of stated GC was observed in bacteriophage DNA derived from the aquatic environment near hospital (Yang et al., 2018) and in chicken feces (Yang et al., 2020). Furthermore, a novel class 1 integron containing *aac(6')-Ib-cr* was revealed in *Salmonella enterica* from hospitalized patients with diarrhea (Yao et al., 2021). Moreover, high co-carriage of *sul1* and *qacEΔ1* was observed in the present study with the occurrence GCs conferring resistance to sulfonamides, trimethoprim, and aminoglycosides, which were in line with previous studies (Riano et al., 2006; Taitt et al., 2017; Zeeshan Khan et al., 2018). The absence of *sul1* and *qacEΔ1* might result in the loss of the 3'-conserved segment of class 1 integron as previously reported (Sáenz et al., 2010). The occurrence of positive and negative combinations of *qacEΔ1* and *sul1* might be a factor involved in the prevalence of MDR. Hence, the present study was showing a high prevalence of GCs and ARGs, which were previously observed in different reservoirs including livestock, poultry, aqua environment, as well as in hospital settings. It was indicated that ducks could also act as a reservoir and took part in the dissemination of ARGs, posing threat to the environmental safety and public health. Thus, continuous surveillance and monitoring is required to understand the complex mechanism of the spread of AMR.

The MGEs such as transposons and insertion sequences are discrete DNA fragments that can move along with ARGs to new locations randomly in the same or other DNA molecules in a particular cell (Partridge et al., 2018). The occurrence of MGEs was found to be on the rise in the present study (Table 3), as recently relatively low percentages

of MGEs such as *trbC* (47.6%), *tnpA*/Tn21 (35.7%), *ISEcp1* (15.5%), *IS26* (6.0%) and only 35 different combinations were recorded in captive giant pandas (Zhu et al., 2020). MGEs might enhance the replication as well as horizontal transfer of ARGs in bacteria, particularly in areas having high biomass (Jia et al., 2017; Chen et al., 2021). Recently, co-occurrence of MGEs and ARGs was noted in the lake sediments, where 43 different types of MGEs were observed with predominance by *tnpA*, *IS91*, *istB* and *int1*. Moreover, the redundancy analysis revealed many MGEs played essential roles in shaping the ARG profiles (Chen et al., 2021). The bacterial community in tap water was analysed from 13 cities along the middle and lower ranges of Yangtze River Basin of China, where significant correlation of *int1* was found with *aadA*, *sul1*, *sul2*, and *tnpA* (Zhang et al., 2021b). In addition, *ISEcp1*, one of the important MGE was found in 38.2% isolates of this study, which might mobilize downstream located ARGs (Zhao and Hu, 2013). In the present study, many strains were contained more than one insertion sequence element in a single gene complex, as previously reported (Bae et al., 2008). Such heterogeneity might be elucidated by continuous recombination and interchange of GCs, meaning the refined genetic rearrangement strategies, which were acquired by organisms to distribute ARGs (Zhao and Hu, 2013). In the present investigations, all Tn7 transposition genes including *tnsA*, *tnsB*, *tnsC*, *tnsD*, and *tnsE* were detected in *E. coli* strains with class 1 and/or class 2 integrons (Table 3), which were previously reported in class 2 integrons (Ramírez et al., 2010; Alonso et al., 2018). Moreover, strains containing Tn7 transposition genes were conferring resistance to trimethoprim, which was in line with a previous study (Orman et al., 2002).

There are complex mechanisms involved in the spread of AMR, and HGT is a well-known phenomenon for the transmission of ARGs mediated by MGEs and integrons within and between bacteria. The spreading of MGEs in the commensal intestinal bacteria was expected in present study, while APEC-linked serotypes were also observed from these diseased ducks. Our findings support the diversity of integrons-GCs and high prevalence of MGEs in diseased ducks, subsequently making it difficult to treat resistant infections in China. Therefore guidelines, procedures and interventions are urgently required to limit antimicrobial use and control the current situation.

5. Conclusion

We systematically detected the diverse and high occurrence of integrons-GCs and MGEs in *E. coli* isolated from intestinal contents or pericardial effusion of diseased ducks from 2018 to 2020 in Sichuan province, China. Our data was illustrated a widespread occurrence of integrons, GCs and MGEs. Further, resistance phenotypes to SXT and C were found significantly associated with the existence of integrons. The *E. coli* strains isolated from the intestinal contents of diseased ducks were carried a high proportion of integrons and mobile components. Subsequently, the high carriage of integrons and MGEs might be involved in the transfer of ARGs to commensal bacteria, risking to public health and environmental safety. Monitoring the occurrence of ARGs, integrons and MGEs are important to understand the complex mechanism of AMR. For the protection of public health, harmonious ecological environment, animal health and welfare, it might help to establish intervention measures for the prevention and control of AMR in duck farms in China.

Ethics approval

The study was approved by the Committee of Experiment Operational Guidelines and Animal Welfare (Licence: SYXK 2019-187) of Sichuan Agricultural University. Moreover, this work did not harm any healthy animal and prior to the collection of intestinal contents/pericardial effusion samples from dead ducks, permission was granted by the ethical committee.

CRedit authorship contribution statement

Shaqiu Zhang: Conceptualization, Investigation, Methodology, Writing - original draft, Writing - review & editing, Funding acquisition; **Shuling Chen:** Investigation, Methodology, Visualization; **Muhammad Abbas:** Writing - original draft, Data curation; **Mingshu Wang:** Data curation, Funding acquisition; **Renyong Jia:** Formal analysis; **Dekang Zhu:** Project administration; **Shun chen:** Validation; **MaFeng Liu:** Validation; **Xinxin Zhao:** Data curation; **Ying Wu:** Data curation; **Qiao Yang:** Data curation; **Juan Huan:** Project administration; **Xumin Ou:** Data curation; **Sai Mao:** Methodology; **Qun Gao:** Methodology; **Di Sun:** Conceptualization; **Bin Tian:** Conceptualization; **Anchun Cheng:** Writing - review & editing, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Conflicts of interest

The authors declare no competing interests.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2021.112475.

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